Asad Hayat

Ph.D Control Science and Engineering | asadhayat@stu.tyust.edu.cn
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Research Interests

Surrogate-assisted evolutionary algorithms; large-scale expensive optimization; machine learning and deep learning; computational intelligence; data science.

Education

Ph.D Control Science — Taiyuan University of Science and Technology, China *2023/10 – 2027/07*
Research on surrogate-assisted evolutionary algorithms for large-scale expensive optimization

MS Computer Science — Capital University of Science and Technology, Pakistan *2018/09 – 2021/04*
Thesis: *A Load-Balanced Task Scheduler for Heterogeneous Systems Based on Machine Learning*

BS IT — IBMS, The University of Agriculture Peshawar, Pakistan *2014/01 – 2017/12*
Final Year Project: *MS Management System* — a Java and SQL-based desktop application

Research Experience

Research Associate — Data Science and Computational Intelligence Lab, Taiyuan University of Science and Technology *2023/10 – present*

- Research on surrogate-assisted single- and multi-objective evolutionary optimization for large-scale expensive problems.
- Developed two novel frameworks:
  ALSO-PMPI (PCA + multi-restart PSO infill criterion, DOCS 2026)
  SASO-AIS (FPCA + GP-guided DE + adaptive infill sampling, submitted to *Knowledge-Based Systems*).

Research Assistant — CORDE Lab, Capital University of Science and Technology *2019/09–2021/06*

- Analyzed CPU/GPU performance on OpenCL applications; designed a machine-learning scheduler for mapping applications to heterogeneous devices.

Work Experience

Lecturer in Computer Science — Lahore Leads University, Lahore, Pakistan *2021/11–2023/09*
Courses: *Introduction to Computing*; *Introduction to Data Science*

Publications

Journal papers

1. **Hayat, A.**, Sun, C., Wu, G., & Hasnain, M. "Surrogate-Assisted Subspace Optimization with Adaptive Infill Sampling for Large-Scale Expensive Optimization." *Submitted: Knowledge-Based Systems*, 2026.
2. Zhao, K., Wang, X., Sun, C., Jin, Y., & **Hayat, A.** "Efficient large-scale expensive optimization via surrogate-assisted subproblem selection." *IEEE Transactions on Evolutionary Computation*, 29(5), pp. 2145–2157, 2025.
3. Liu, X., Sun, C., Jin, Y., & **Hayat, A.** "Expensive many-objective evolutionary optimization with a self-adaptive surrogate model." *IEEE Transactions on Emerging Topics in Computational Intelligence*, 10(1), pp. 348–357, 2025.

4. Hussain, A., **Hayat, A.**, & Hasnain, M. "Using hate speech detection techniques to prevent violence and foster community safety." *Journal of Artificial Intelligence*, 7(1), p. 485, 2025.
5. Hasnain, M., Ghani, I., Smith, D., Asaf, M., Saleem, S., **Hayat, A.**, & Jeong, S. R. "Recurrent Neural Network (RNN)–based approach to uncover the relationship between block size and blockchain performance." *Applied Computational Intelligence and Soft Computing*, 2024(1), 2024.
6. Nadir, M. N., Rathore, M. S., **Hayat, A.**, & Mansoor, J. A. "CPU vs. GPU: Performance comparison of OpenCL applications on a heterogeneous architecture." *Journal of Computing & Biomedical Informatics*, 7(2), pp. 1–14, 2024.
7. Hasnain, M., **Hayat, A.**, & Hussain, A. "Revolutionizing chronic obstructive pulmonary disease care with the Open AI application: ChatGPT." *Annals of Biomedical Engineering*, pp. 2100–2102, 2023.
8. **Hayat, A.**, Khalid, Y. N., Rathore, M. S., & Nadir, M. N. "A machine learning-based resource-efficient task scheduler for heterogeneous computer systems." *The Journal of Supercomputing*, 79(14), pp. 15700–15728, 2023.
9. Sajawal, M., Usman, S., Alshaikh, H. S., **Hayat, A.**, & Ashraf, M. U. "A predictive analysis of retail sales forecasting using machine learning techniques." *Lahore Garrison University Research Journal of Computer Science and Information Technology*, 6(04), pp. 33–45, 2022.

Conference papers

10. **Hayat, A.**, & Sun, C. "Adaptive Latent Space Optimization via PCA and Multi-Restart PSO-Based Infill Criterion for Large-Scale Expensive Problems." *8th International Conference on Data-driven Optimization of Complex Systems (DOCS)*, 2026.
11. Aamir, A., Wang, X., Rehman, M. U., **Hayat, A.**, Muhammad, T., & Tauqeer, A. "Design and implementation of PLC-based automated vegetable cutting system for enhanced food processing efficiency." *International Conference on Engineering and Emerging Technologies (ICEET)*, pp. 1–6. IEEE, 2025.
12. Muhammad, T. F., Muhammad, N. N., & **Hayat, A.** "BAP: Innovative approach to integrate machine learning for researcher's ranking through parameters assessment." In *Proc. of the 2025 6th International Conference on Advancements in Computational Sciences (ICACS)*, pp. 1–7. IEEE, 2025.
13. **Hayat, A.**, Nadir, M. N., & Rathore, M. S. "A load-balanced task scheduler for heterogeneous systems based on machine learning." *1st International Conference on Computing Research (ICCoR 2022)*.
14. Nadir, M. N., **Hayat, A.**, & Rathore, M. S. "Analysis of OpenCL applications on CPU and GPU." *1st International Conference on Computing Research (ICCoR 2022)*.

Languages

Programming: Python, MATLAB, Java, C/C++

Languages: English: fluent; Urdu: native; Chinese: HSK Level 3 passed

References

Prof. Chaoli Sun — Ph.D Supervisor
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Dr. Muhammad Hasnain — Collaborator & Former Colleague
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